1.1 Physical Quantities & Units

Question Paper

Course	CIE A Level Physics
Section	1. Physical Quantities & Units
Topic	1.1 Physical Quantities & Units
Difficulty	Easy

Time allowed: 20

Score: /10

Percentage: /100

Question 1

Which of the following is an SI base unit?

A. Current

B. Gram

C. Kelvin

D. Volt

[1 mark]

Question 2

Which definition is correct and uses only quantities rather than units?

A. Density is mass per cubic metre

B. Potential difference is energy per unit current

C. Pressure is force per unit area

D. Speed is distance travelled per second

[1 mark]

Question 3

Five energies are listed below.

5kJ

5mJ

5MJ

5nJ

Starting with the smallest first, what is the order of increasing magnitude of these energies?

A. 5kJ → 5mJ → 5MJ → 5nJ

B. 5nJ → 5kJ → 5MJ → 5mJ

C. 5nJ → 5mJ → 5kJ → 5MJ

D. 5mJ → 5nJ → 5kJ → 5MJ

[1 mark]

Question 4

What is a reasonable estimate of the average gravitational force acting on a fully grown woman standing on the Earth?

A. 60 N

B. 250 N

C. 350 N

D. 650 N

[1 mark]

Question 5

Which product-pair of metric prefixes has the greatest magnitude?

A. pico × mega

B. nano × kilo

C. micro × giga

D. milli × tera

[1 mark]

Question 6

Which pair contains one vector and one scalar quantity?

A	Displacement	Acceleration
B	Force	Kinetic energy
C	Momentum	Velocity
D	Power	Speed

[1 mark]

Question 7

Which statement includes a correct unit?

A. Energy = 7.8 Ns

B. Force = 3.8 Ns

C. Momentum = 6.2 Ns

D. Torque = 4.7 Ns

[1 mark]

Question 8

Which quantity can be measured in electronvolts (eV)?

A. Electric charge

B. Electric potential

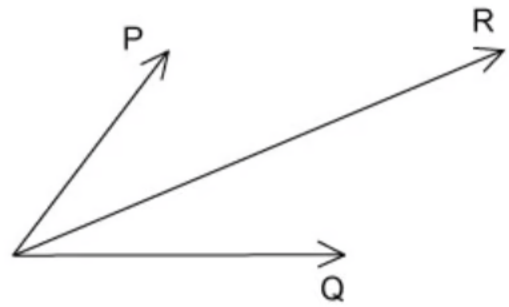
C. Energy

D. Power

[1 mark]

Question 9

Two physical quantities **P** and **Q** are added together. The sum of **P** and **Q** is **R**, as shown in the diagram.



Which quantity could be represented by **P** and by **Q**?

A. Kinetic energy

B. Power

C. Speed

D. Velocity

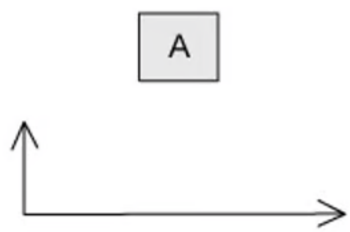
[1 mark]

Question 10

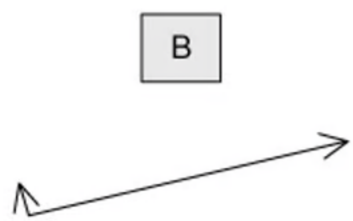
The arrow represents the vector **R**.

Which diagram does **not** represent **R** as two perpendicular components?

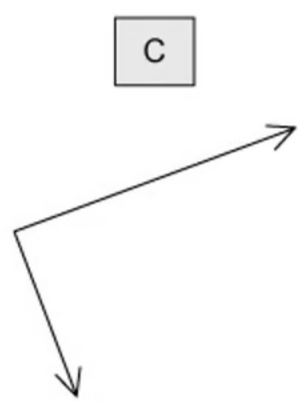
A.



B.



C.



D.

[1 mark]

Model Answers: Easy

1

The correct answer is **C** because:

- The main SI units you need to be aware of are:
  - Metre (m)
  - Kilogram (kg)
  - Second (s)
  - Ampere (A)
  - Kelvin (K)
  - Mole (mol)
- Kelvin is the only SI unit listed

A is incorrect as current is a quantity, not a unit

B is incorrect as the correct SI unit for mass is the kilogram

D is incorrect as the volt is a derived unit

2

The correct answer is **C** because:

- The question asks you to select the correct statement which uses quantities only
- Force and area are both quantities, not units

A is incorrect as while the statement is correct, a metre is a unit

B is incorrect as the statement is incorrect, potential difference is the energy per unit charge

D is incorrect as while the statement is correct, second is a unit

3

The correct answer is **C** because:

- If we convert each power of ten we get:
  - 5 kJ = 5×10^3 J
  - 5 mJ = 5×10^{-3} J
  - 5 MJ = 5×10^6 J
  - 5 nJ = 5×10^{-9} J
- Therefore, from smallest to largest: **5nJ → 5mJ → 5kJ → 5MJ**

4

The correct answer is **D** because:

- The average mass of an adult woman is about 65 kg
- Rounding up, g is about 10 N kg^{-1} on Earth
- To find weight we use $W = mg$
- So, $W = 65 \times 10 = 650$ N

5

The correct answer is **D** because milli × tera is equal to $10^{-3} \times 10^{12} = 10^9$

A is incorrect as this is equal to $10^{-12} \times 10^6 = 10^{-6}$

B is incorrect as this is equal to $10^{-9} \times 10^3 = 10^{-6}$

C is incorrect as this is equal to $10^{-6} \times 10^9 = 10^3$

6

The correct answer is **B** because:

- Force is a vector quantity as it has magnitude and direction
- Kinetic energy is a scalar quantity as it has only magnitude

A is incorrect as displacement and acceleration are both vector quantities

C is incorrect as momentum and velocity are both vector quantities

D is incorrect as power and speed are both scalar quantities

7

The correct answer is **C** because change in momentum (or impulse) is equal to force × time, giving a unit of *Ns*

A is incorrect as the unit of energy is *J* or $kg\ m^2 s^{-2}$ or *N m*

B is incorrect as the unit of force is *N* or $kg\ m\ s^{-2}$

D is incorrect as the unit of torque is *N m* or $J\ rad^{-1}$ or $kg\ m^2 s^{-2}$

8

The correct answer is **C** because:

- The definition of an electron volt is the energy that an electron gains when it travels through a potential of one volt
- 1 eV is equal to 1.6×10^{-19} J

9

The correct answer is **D** because:

- The diagram represents a quantity which has a magnitude and direction, so it must be a vector
- Velocity is the only vector quantity listed

A, B and **C** are incorrect as kinetic energy, power and speed are all scalar quantities as they all have a magnitude but no direction

10

The correct answer is **C** because:

- The vector sum of the two perpendicular components should provide a resultant vector equal to the vector **R**
- If we complete the diagrams, we can see that option C is the odd one out as it produces a resultant vector in a downward direction, therefore it cannot be equal to vector **R**